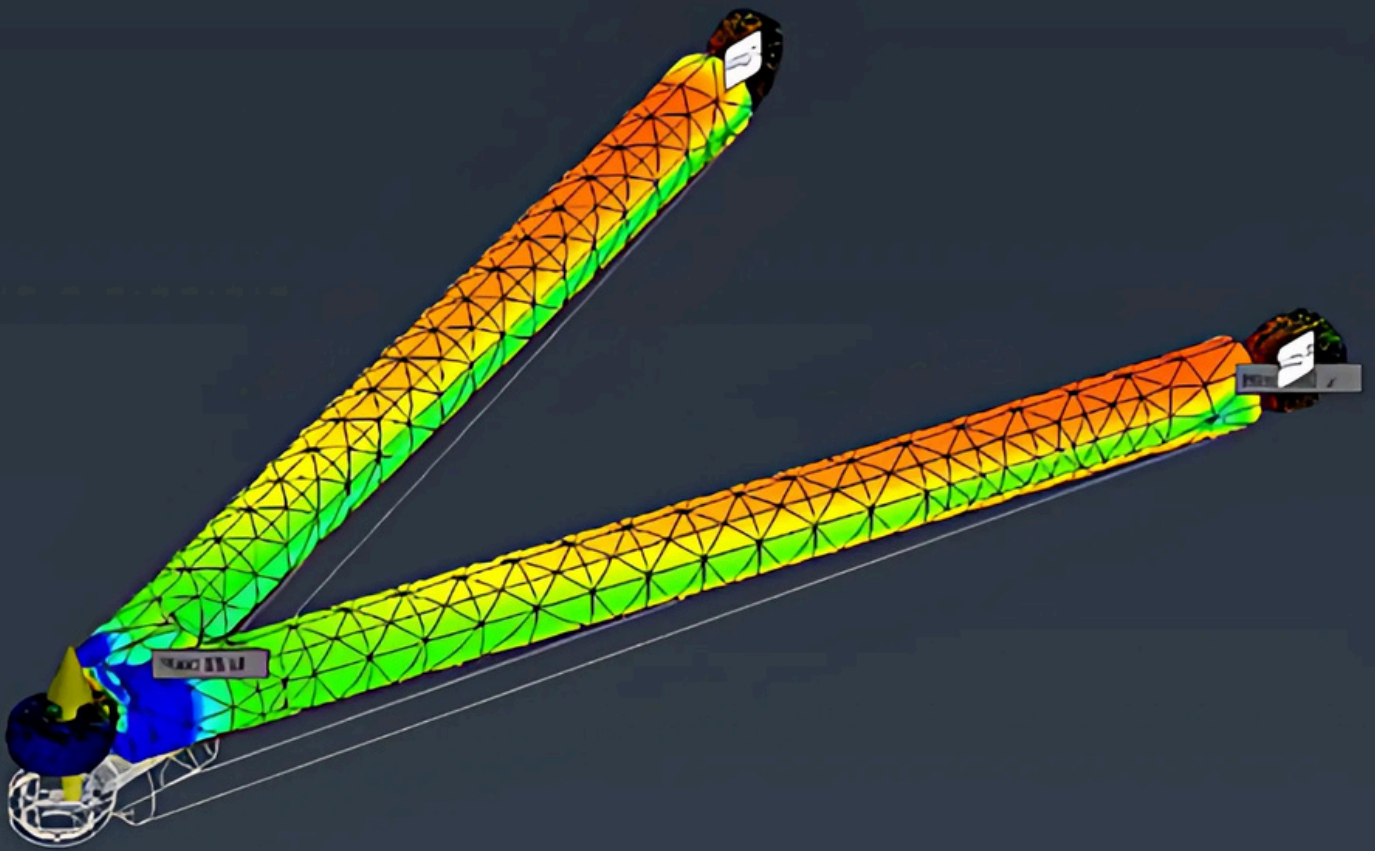


OCTOBER 2024



CONTROL ARMS

CAN THEY TAKE THE
FORCES WITHOUT
FAILING?

EXPANDING THE TEAM

LEARN ABOUT THE
HIRING CAMPAIGN,
REORGANIZATION AND
MORE!

WHAT CAN YOU EXPECT?

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WHAT IS AUS RACING
- 04** | WHAT ARE THE
STRESSES ON CONTROL ARMS
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BRAKING WITH EBD
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HOW YOU CAN SPONSOR US

WHAT IS AUS RACING

AUS Racing is the Formula Student team from the American University of Sharjah. Formula Student is an international engineering design competition that graciously provides a platform to aspiring engineers to build an open-wheel, single seater racecar and simulate all the different nuances that culminate to account for its structural integrity, performance, economic feasibility and industrial standards.

Students compete in different static and dynamic events to assess the car, thus testing the engineering design, management skills and the overall team spirit of the contesting teams. It is an esteemed competition that unlocks new horizons for AUSers

to venture into as we progress into a new era of technological competence and innovation.

Here at AUS Racing, we hope to open up opportunities for students for the foreseeable future to take part in a project that will teach them engineering, management and budgeting. In essence, real world engineering.

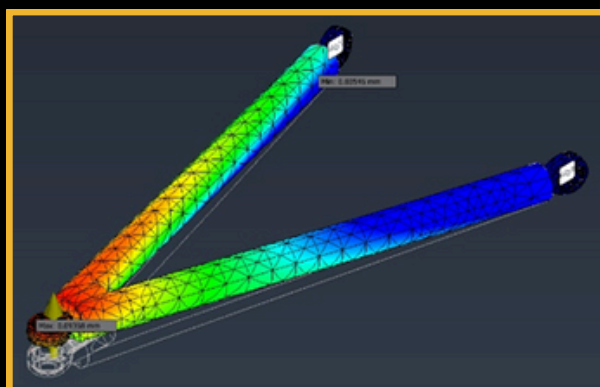
STRESSES ON CONTROL ARMS

Using the initial design for the control arms, the team was able to run some initial simulations for validating calculations and design choices.

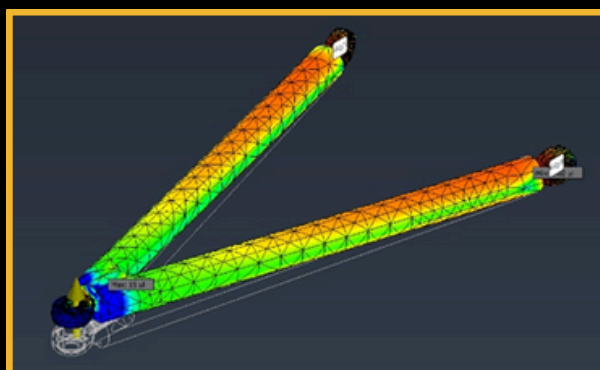
The colours are not just for aesthetics. They allow analysts or engineers to understand how a specific part responds to different forces or loads on them.

Simulations can give a lot of data such as the stress distribution, required safety factor, breaking point etc. Our control arms have only been run through preliminary simulations.

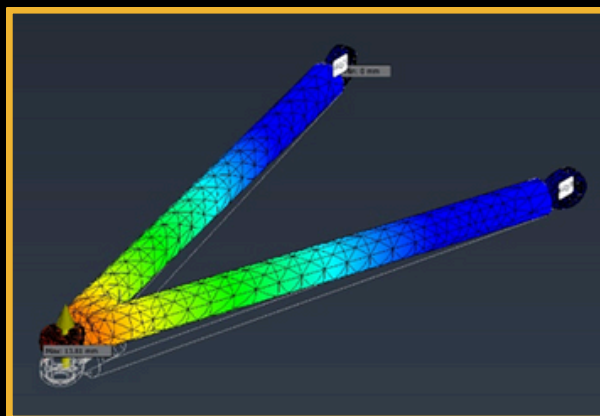
As more designs are complete, better and more informative simulations will be run through Ansys', our sponsor's, software.



Principle stress

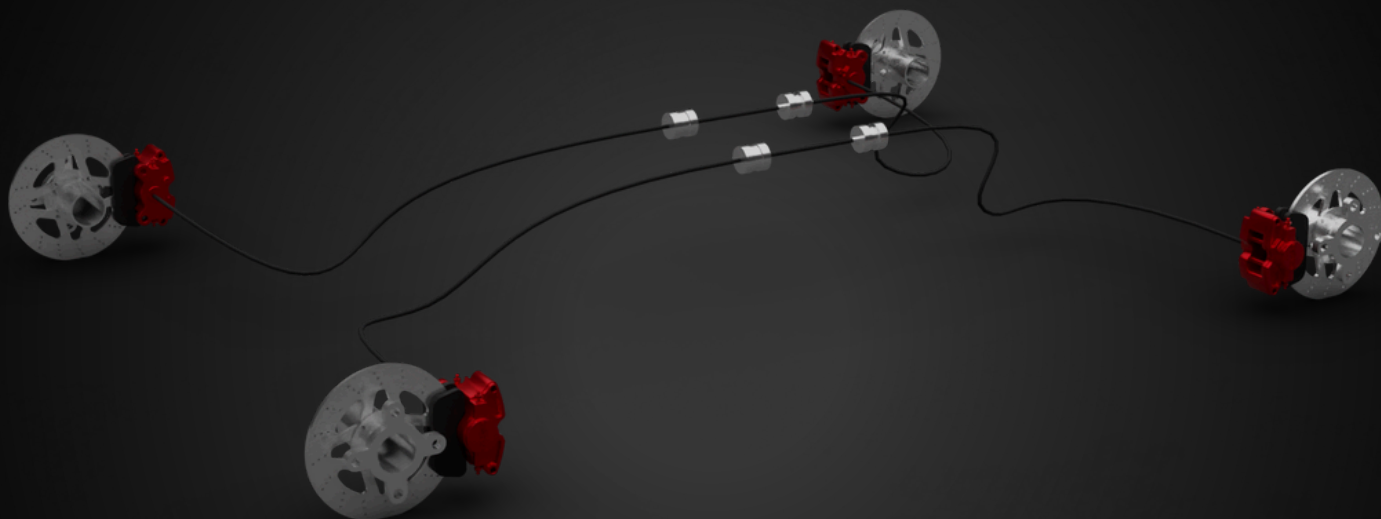


Stress in XX



Safety Factor

BRAKING WITH EBD



EBD stands for Electronic Brakeforce Distribution. EBD is a system used in every commercial vehicle for improving braking performance and safety.

For the most efficient deceleration, all tires need to slow down and stop at the same time.

Cars which are heavier towards the front generally have larger brakes in the front as they require

more force to decelerate at the same rate as the rear brakes.

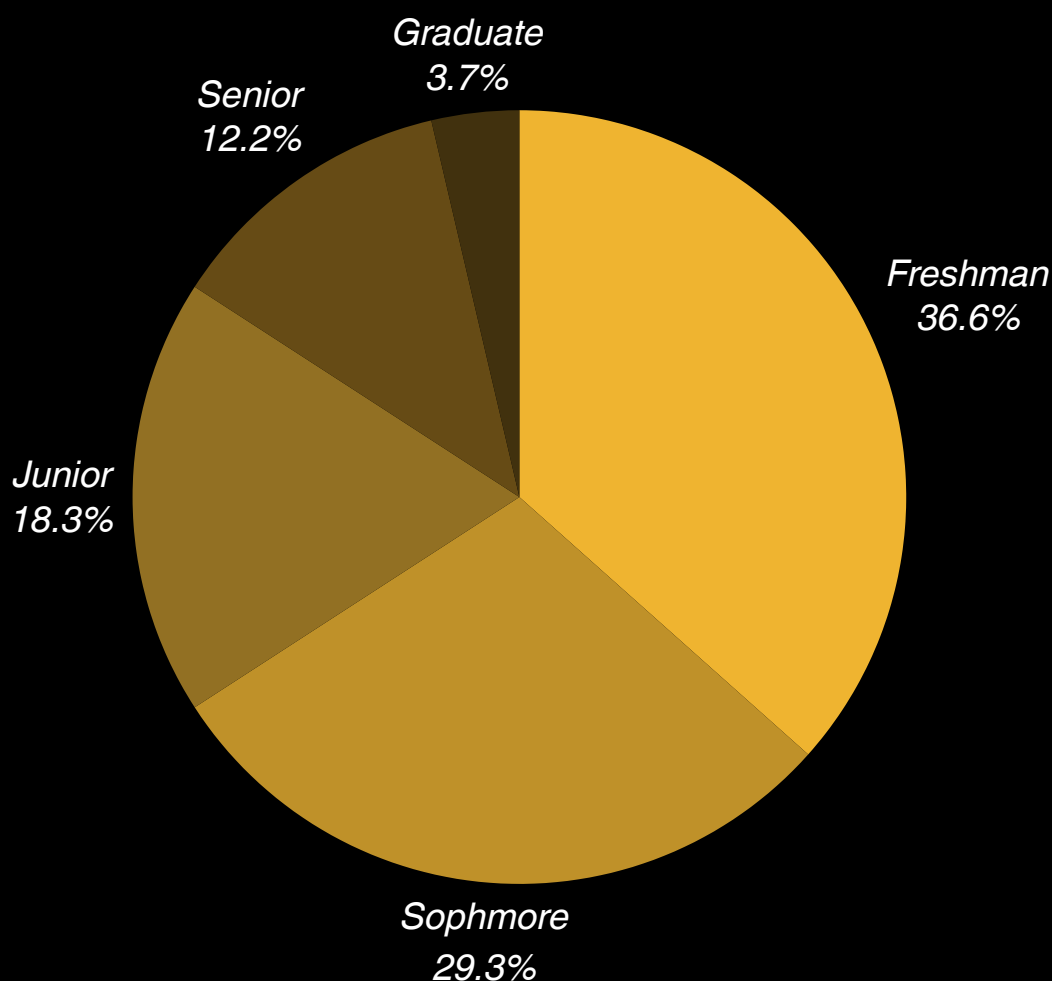
However, when brakes are applied, depending on the magnitude, the car pitches forward. Due to this, the weight on the wheels are never the same.

EBD takes this dynamic loading into consideration and accurately changes how much brakeforce goes to the front and rear wheels.

OUR HIRING CAMPAIGN

With the growing need for a larger team due to the increased workload, applications were opened for two work weeks. We received applications from over 80 students distributed over all of AUS' colleges and 18 majors.

The distribution of the standings of our applicants is illustrated in the pie chart. A distribution like this allowed the team to shortlist and recruit applicants from every standing more easily and effectively.



MEET THE MAIN BOARD

This month marked a major milestone for the team as the organization was restructured. A variety of new roles were introduced along with the main board of management.

This new organization is crucial for streamlining the team's workflow. With the main board in place, there have already been major improvements in overall team performance and progress.



Saurav Gupte
Vice-Principal



Saad Ibrahim
Team Principal



Earielle Capili
Executive Assistant



Shwetambari Abhirajan
External Coordinator



Ali Qayed Ahli
Head of Engineering



Mohammad Shaikh
Internal Coordinator

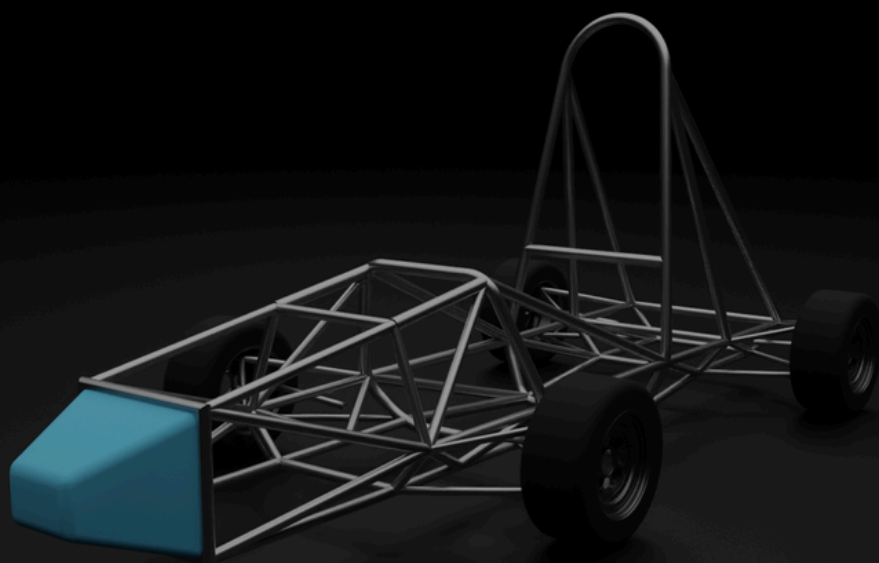
HOW YOU CAN SPONSOR US

What if you were at the forefront of engineering innovation?

What if your company was responsible for bringing our aspiring engineers' ideas to life?

What if your business could get its message across the world through Formula Student?

If your answer to any of these questions is yes, click [here](#) and check out what we have in store for our generous sponsors and learn how to become one yourself!



AUS

American
University
of Sharjah

GOLD

Ansys

BRONZE

AUTOMECH
GROUP